

Curriculum Vitae

Jae Hak Son

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Academic Position

- 2021 - Current Rutgers University Postdoctoral Associate
Advisor: Dr. Christopher E. Ellison
- 2019 - 2021 Yale School of Public Health Postdoctoral Associate
Advisor: Dr. Serap Aksoy

Education

- 2019 University of Houston Ph.D. in Biology and Biochemistry
Supervisor: Assistant Professor Richard P. Meisel
- 2013 Seoul National University M.S. in Biological Science
Supervisor: Associate Professor Piotr G. Jablonski
- 2011 Hanyang University B.S. (Honor) in Molecular Life Science
Supervisor: Professor Young Gyu Chai

Academic Activities

Publications

11. Li X, Visser S, Son JH, Geuverink E, Kivanç EN, Wu Y, Schmeing S, Pippel M, Anvar SY, Schenke MA, Marec F, Robinson MD, Meisel RP, Wimmer E, Zande L, Bopp D, Beukeboom LW (2024). Divergent evolution of male-determining loci on different proto-Y chromosomes of the housefly. *Nature Communications*, 15(1): 5984
10. Saarman NP, Son JH, Zhao H, Cosme LV, Kong Y, Li M, Wang S, Weiss BL, Echodu R, Opiro R, Serap Aksoy S, and Caccone A (2023). Genomic evidence of sex chromosome aneuploidy and infection-associated genotypes in the tsetse fly *Glossina fuscipes*, the major vector of African trypanosomiasis in Uganda. *Infection, Genetics and Evolution*, 105501
9. Son JH[†], Weiss BL[†], Schneider DI, Dera KM, Gstöttenmayer F, Opiro R, Echodu R, Saarman NP, Attardo GM, Onyango M, et al. (2021) Infection with endosymbiotic *Spiroplasma* disrupts tsetse (*Glossina fuscipes fuscipes*) metabolic and reproductive homeostasis. *PLoS pathogens*, 17(9): e1009539. [†]equal contribution
8. Yang L, Weiss BL, Williams AE, Aksoy E, de Silva Orfano A, Son JH, Wu Y, Vigneron A, Karakus M, Aksoy S (2021) Paratransgenic manipulation of a tsetse microRNA alters the physiological homeostasis of the fly's midgut environment. *PLoS pathogens*, 17(6): e1009475.
7. Adhikari K, Son JH, Rensink AH, Jaweria J, Bopp D, Beukeboom LW, Meisel RP (2021). Temperature-dependent effects of house fly proto-Y chromosomes on gene expression could be responsible for fitness differences that maintain polygenic sex determination. *Molecular Ecology* 30(22), 5704-5720
6. Son JH^{*}, Meisel RP^{*} (2020) Gene-level, but not chromosome-wide, divergence between a very young house fly proto-Y chromosome and its homologous proto-X chromosome. *Molecular Biology and Evolution*, 38(2): 606-618 ^{*}correspondence
5. Son JH, Kohlbrenner T, Heinze S, Beukeboom LW, Bopp D, Meisel RP (2019) Minimal effects of proto-Y chromosomes on house fly gene expression in spite of evidence that selection maintains stable polygenic sex determination. *Genetics*, 213: 313-327
4. Meisel, RP, Davey T, Son JH, Gerry AC, Shono T, Scott JG (2016) Is Multifactorial Sex Determination in the House Fly, *Musca domestica* (L.), Stable Over Time? *Journal of Heredity*, 107(7): 615-625
3. Yang E, Son JH, Lee SI, Jablonski PG, Kim HY (2016) Water striders adjust leg movement speed to optimize takeoff velocity for their morphology. *Nature Communications*, 7(1), 13698

2. Koh JS, Yang E, Jung GP, Jung SP, Son JH, Lee SI, Jablonski PG, Wood RJ, Kim HY, Cho KJ (2015) Jumping on water: Surface tension-dominated jumping of water striders and robotic insects. *Science*, 349(6247): 517-521

1. Son JH, Han CS, Lee SI, Jablonski PG (2014) Effect of sex, hunger and relative body size on the use of ripple signals in the interactions among water striders *Gerris latiaabdominis*. *Journal of Asia-Pacific Entomology*, 17(4): 653-658

Preprints

Son JH, Lawlor MA, Virani M, Cao W, Levine MT, Ellison C. (2025). Convergence and conflict among telomere-specialized transposons across 60 million years of *Drosophilid* evolution. *bioRxiv* (in 2nd revision, April 2026, *Genome Research*)

In preps

Son JH, Luecke D, Bista B, Cao W, Beukeboom LW, Bopp D, Ellison CE, Saelao P, Meisel, RP (in prep) The house fly polygenic sex determination system provides insight into the early stages of sex chromosome differentiation

Son JH, Volski M, Cao W, Ellison CE. (In prep) Evolutionary dynamics of the nuclear export factor gene family across the *Drosophila* genus

Invited Presentations

- 2026 The Plant and Animal Genomics Conference, San Diego, CA, USA
- 2023 Agriculture and Life Sciences, Seoul National University, Seoul, Korea
- 2016 XXV International Congress of Entomology, Orlando, FL, USA.

International / National / Regional Conference Oral Presentations

- 2026 New York Population Genetics Meeting, New York, USA
- 2024 The Allied Genetics Conference, Metro Washington, DC, USA
- 2022 Department of Genetics, Human Genetics Institute of New Jersey, Rutgers University, Piscataway, NJ, USA
- 2018 Southeast Texas Evolutionary Genetics and Genomics Symposium, Houston, TX, USA
- 2018 Houston Regional Ecology and Evolution Student Symposium, Houston, TX, USA
- 2017 9th Annual Graduate Student Symposium, Houston, TX, USA
- 2017 Houston Regional Ecology and Evolution Student Symposium, Houston, TX, USA
- 2016 8th Annual Graduate Student Symposium, Houston, TX, USA
- 2015 Houston Regional Ecology and Evolution Symposium, Houston, TX, USA.
- 2013 Korean Society of Applied Entomology Regular Conference, Muju, Korea.

International / National / Regional Conference Poster Presentations

- 2025 66th Annual Drosophila Research, San Diego, CA, USA
- 2022 Transposable Elements, Cold Spring Harbor, NY, USA
- 2022 63th Annual Drosophila Research, San Diego, CA, USA. Online
- 2020 The Allied Genetics Conference, Online
- 2019 New England Association of Parasitologists, New Haven, CT, USA
- 2019 60th Annual Drosophila Research, Dallas, TX, USA
- 2018 West Gulf Coast Regional Conference (Korean-American Bio-Medical Symposium), Houston, TX, USA
- 2018 Population, Evolutionary, Quantitative Genetics Conference, Madison, WI, USA
- 2017 West Gulf Coast Regional Conference (Korean-American Bio-Medical Symposium), Houston, TX, USA
- 2017 The annual meeting of the Society for Molecular Biology and Evolution 2017, Austin, TX, USA
- 2014 6th Annual Graduate Student Symposium, Houston, TX, USA

- 2012 14th International Behavioral Ecology Congress, Lund, Sweden.
- 2011 66th Annual Meeting of the Korean Association of Biological Sciences, Chuncheon, Korea.

Research Experience (visiting and assistant)

- 2016 Visiting student, Institute of Molecular Life Science, University of Zurich
(Principal investigator: Dr. Daniel Bopp)
- 2010 Research Assistant, Lab of Behavioral Ecology & Evolution, Seoul National University
- Project: Soft morphing locomotion mechanism of water striders
- 2009 Research Assistant, Lab of Behavior & Ecology, Ewha Woman University
- Project: Acoustic communication in Birds, tit family.

Grant

- 2018 Sigma Xi Grants-in-Aid of Research

Awards and Fellowships

- 2019 Cullen Fellowship Travel Grant, University of Houston
- 2018 Departmental Student Travel Award, University of Houston
- 2017 Cullen Fellowship Travel Grant, University of Houston
- 2016 Departmental Student Travel Award, University of Houston
- 2016 Travel Stipend for the 2016 International Congress of Entomology
(ICE Program Enhancement Funds)
- 2014-2019 Graduate Tuition Fellowship, University of Houston
- 2014-2016 Presidential Cullen and Stella Ehrhardt Fellowship, University of Houston
- 2011-2012 Research Assistant Scholarship, Seoul National University
- 2010 Merit-based Scholarship, Hanyang University
- 2009 Mentorship Scholarship, Hanyang University
- 2005 Merit-based Scholarship, Hanyang University
- 2005 Student Service Scholarship, Hanyang University

Instructional Experience

- 2011, 2012 Teaching assistant, Laboratory of Biology
- 2014 - 2019 Teaching assistant, Laboratory of Biology, Genetics, Human Anatomy

Skills and Proficiencies

- Bioinformatics and Data Analytics: R, Python, Perl, Unix (High-Performance Computing environment), Reproducible workflow orchestration via Snakemake, Conda, and Containers
- Molecular Biology: DNA-seq (Illumina, Oxford Nanopore), RNA-seq (Illumina), small RNA-seq (illumine), microinjection for RNA interference (RNAi), DNA/RNA extractions, PCR, RT-qPCR, Bacterial transformations, and cloning
- Cytogenetics: Fluorescence in situ hybridization (FISH)
- Classical Genetics: Backcrossing with chromosomal phenotypic markers

Community and Outreach Activities

Peer Review

- 2023 in Communication Biology, Genetics
- 2021 in Genomics

- 2018 in Molecular Biology and Evolution
- 2018 in Journal of the Kansas Entomological Society

Membership of Professional Societies

Genetics Society of America

Society of Molecular Biology and Evolution

Student Mentorship

2023-2024	Justin Seo	Dr. Christopher Ellison lab, Rutgers University
2022	Mikaela Benkert	New Jersey STEM Network, Montclair State University
2017-2019	Dalia Aldin	Dr. Richard Meisel lab, University of Houston

Public Outreach

2016 - 2018	Volunteering supervisor, National Math and Science Competition hosted by Korean-American Scientists and Engineers Association (KSEA), South Texas/Houston Chapter, Houston, TX
2025 - current	Outreach Director, Korean-American Scientists and Engineers Association (KSEA) New Jersey Chapter, NJ